THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 7: **Temperature and its Measurement** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Why are bakers' gloves made thick and of cloth? 1

(a) to resist heat conduction **(b)** to be heavy
(c) to look good **(d)** to conduct

2. If you want to cool a hot drink faster, which is better? 1

(a) wrap in cloth **(b)** cover it
(c) keep it in a thermos **(d)** pour it into a wide steel plate

3. The thermometer tube is narrow so that: 1

(a) a small expansion makes a large change in length
(b) it is clear
(c) it is cheap
(d) it is light

4. Why do the metal parts of a bicycle feel cold on a winter morning? 1

(a) it is wet **(b)** metal is colder than the air
(c) metal conducts heat away from the hand **(d)** metal makes cold

5. The reason why a fever reading in mouth and armpit differs slightly is: 1

(a) mercury changes **(b)** body parts differ in temperature
(c) the scale is wrong **(d)** air is hot

6. Why is the part of the clinical thermometer scale more finely divided in tenths of a degree? 1

(a) small changes in fever matter **(b)** easy printing
(c) cheap **(d)** for decoration

7. Why is food kept in a closed container warmer for longer? 1

(a) it is metal **(b)** air gets trapped and heat loss is less
(c) food produces heat **(d)** the container is hot

8. Which of these cannot be an accurate method to compare two temperatures? 1

(a) using a thermometer **(b)** using infrared thermometer
(c) touching with hand **(d)** using digital thermometer

9. Which factor does not affect the rate of heat transfer from a hot cup? 1

(a) the temperature difference **(b)** the material of cup
(c) the surface area **(d)** the colour of the table label

10. Why does a thermometer need time to reach the correct reading? 1

(a) mercury is lazy **(b)** heat takes time to flow into the bulb
(c) the scale moves slowly **(d)** the glass is thick

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are two layers of vests warmer than one thick vest of the same total thickness? 2

12. A clinical thermometer shows 37.5 °C and the doctor says 'slight temperature'. Explain. 2

13. Why does the mercury level fall suddenly when a clinical thermometer is shaken, but not when removed from the mouth? 2

14. Why is a steel tiffin not good for keeping food hot long? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Why can ice float in a drink and make it cold? 3

16. Explain why the walls of cold storage are made of insulating material. 3

17. Compare the roles of conductor and insulator in an electric kettle. 3

OR

Explain why a thermometer in a bucket of hot water reads higher than the reading of the room before and settles to a reading. 3

18. Why is a mercury thermometer not used in children's mouths nowadays? 3

OR

Describe how you would calibrate your own thermometer using ice and boiling water. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Design a safe plan to teach younger students how to use a clinical thermometer and read it. 5

OR

Describe how to check the boiling point of water with a lab thermometer and discuss why the reading may vary slightly. 5

20. Explain with three examples that temperature is not the same as the amount of heat. 5

OR

Discuss the role of temperature in cooking and in storing food and how to ensure safety. 5

END OF QUESTION PAPER Total: 40 Marks

MASTERMIND